Single-Cell RNA-seq Analysis Reveals T-cell Exhaustion States, Cellular Plasticity, and Tumour Heterogeneity in Melanoma.

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Abstract

Melanoma is a highly heterogeneous malignancy shaped by dynamic interactions between malignant cells and the surrounding immune microenvironment. Understanding this complexity is essential for improving immunotherapy and targeted treatment strategies. In this study, we performed an in-depth exploratory single-cell RNA sequencing (scRNA-seq) analysis of a publicly available melanoma dataset using a fully reproducible Python workflow built with Scanpy. The analysis included preprocessing, dimensionality reduction, graph-based clustering, metadata integration, lineage annotation, gene-signature scoring, and iterative tumour-state refinement. Distinct immune and stromal populations were identified, including T cells, B cells, NK cells, monocytes/macrophages, endothelial cells, platelets, and tumour-associated populations. Focused interrogation of the T-cell compartment revealed heterogeneous exhaustion-associated states characterized by variable expression of PDCD1, LAG3, TOX, and related immune checkpoint markers. Subsequent tumour-focused analyses demonstrated multiple melanoma transcriptional programs, including differentiated melanocytic cells, proliferative subpopulations, stress-responsive cells, and VIM-high plastic states. Importantly, the study also highlights a practical feature of real-world single-cell analysis: initial metadata annotations often require iterative biological validation and refinement. Together, these findings demonstrate how scRNA-seq can resolve clinically relevant cellular ecosystems in melanoma and provide a robust portfolio framework for independent cancer bioinformatics research.

Keywords

Melanoma; single-cell RNA-seq; tumour microenvironment; T-cell exhaustion; tumour heterogeneity; Scanpy; computational biology; cancer genomics.

Abstract

Melanoma is characterized by substantial intratumoral heterogeneity and complex immune interactions within the tumour microenvironment. In this study, we performed

an exploratory single-cell RNA sequencing (scRNA-seq) analysis of a publicly available melanoma dataset using Python and Scanpy. We constructed an end-to-end workflow including preprocessing, dimensionality reduction, clustering, marker-based annotation, gene-signature scoring, and biological interpretation. We identified major cell populations including T cells, B cells, NK cells, macrophages, endothelial cells, CAFs, and malignant cells. Focused analysis of T cells revealed heterogeneous exhaustion-associated states defined by elevated expression of PDCD1, LAG3, TOX, and related markers. Tumour-focused analysis demonstrated multiple melanoma programs, including differentiated melanocytic cells, proliferative cells, stress-responsive cells, and VIM-high plastic states. These results illustrate how single-cell transcriptomics can resolve clinically relevant cellular states in melanoma and provide a reproducible framework for independent cancer bioinformatics research.

Introduction

Melanoma is an aggressive skin cancer with high metastatic potential and marked transcriptional plasticity. Responses to therapy, especially immune checkpoint blockade, are strongly influenced by both tumour-intrinsic states and the composition of the tumour microenvironment. Bulk RNA sequencing averages signals across many cells and cannot fully resolve rare or transitional populations. In contrast, scRNA-seq enables characterization of heterogeneous cell states at single-cell resolution.

Two biological themes are especially relevant in melanoma: (1) dysfunctional or exhausted T-cell states associated with chronic antigen stimulation, and (2) tumour cell plasticity, where melanoma cells transition between differentiated and invasive phenotypes. Here, we applied a single-cell workflow to investigate both immune and tumour compartments in a melanoma dataset.

Materials and Methods

Computational Environment

Analyses were performed in Python using:

- Scanpy
- Pandas
- NumPy
- Matplotlib

- Jupyter Lab

Data Processing

A processed gene-expression matrix and metadata table were loaded and converted into an AnnData object. Cell-level metadata included patient/sample identifiers, malignant annotations, and broad cell-type labels.

Preprocessing Pipeline

The following standard single-cell workflow was applied:

1. Total-count normalization
2. Log-transformation
3. Highly variable gene selection
4. Scaling
5. Principal component analysis (PCA)
6. Nearest-neighbor graph construction
7. UMAP embedding
8. Leiden clustering

Cell-Type Annotation

Clusters were interpreted using known lineage markers and provided metadata labels to identify major populations.

T-cell Exhaustion Scoring

A custom exhaustion score was computed from canonical markers where available, including:

- PDCD1
- LAG3
- TOX
- TIGIT

- HAVCR2
- CTLA4

T cells were subsetted and reclustered to identify functional heterogeneity.

Tumour-State Analysis

Malignant-labeled cells were extracted and iteratively refined using melanoma lineage markers:

- PMEL
- TYR
- DCT
- MITF

Gene expression patterns and cluster markers were used to remove contaminating immune populations and define tumour-specific programs.

Results

Global Tumour Microenvironment Map

Unsupervised clustering identified multiple major cell populations:

- T cells
- B cells
- NK cells
- Macrophages
- Endothelial cells
- CAFs
- Tumour cells

These populations formed distinct structures in UMAP space, consistent with transcriptionally separable lineages.

T-cell Heterogeneity and Exhaustion

T-cell reclustering revealed multiple subpopulations with variable exhaustion scores. Elevated PDCD1, LAG3, and TOX expression was enriched in specific T-cell clusters, indicating heterogeneous dysfunctional states rather than a single uniform T-cell compartment.

Tumour Heterogeneity

Initial malignant annotations contained mixed populations, highlighting the need for iterative refinement. After melanoma-marker filtering and cluster-level validation, tumour cells showed multiple biological programs.

Differentiated Melanocytic State

Many tumour cells expressed lineage markers such as DCT, PMEL, and TYR, consistent with melanocytic identity.

Proliferative State

A subset of cells showed elevated PCNA, suggesting active cell-cycle progression.

Stress-Responsive State

Variable FOS expression suggested immediate early signaling or stress-response activation.

Plastic / Dedifferentiated Tendency

Partial VIM expression indicated a more mesenchymal or plastic phenotype, potentially linked to therapy resistance.

Figure Summary

Figure 1. Global Cell-Type Map

The global UMAP embedding separated major transcriptional populations across the melanoma microenvironment. Distinct clusters corresponding to T cells, NK cells, B cells, monocytes, inflammatory monocytes/macrophages, rare cells, and platelets were observed. The strong separation of these populations supports successful normalization and clustering.

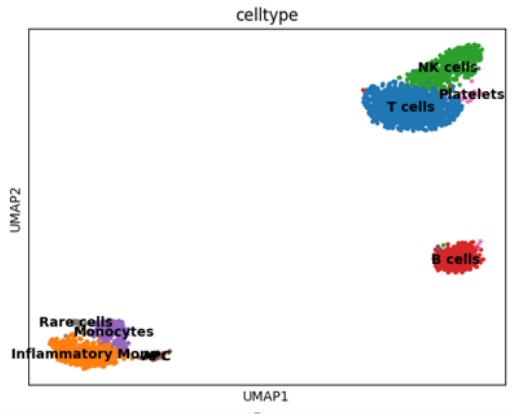


Figure 2. Malignant Annotation Map

Projection of malignant versus non-malignant labels demonstrated that tumour-associated cells occupy a large central manifold, while multiple immune populations remain clearly distinct. A smaller unresolved fraction suggests transitional or uncertain states.

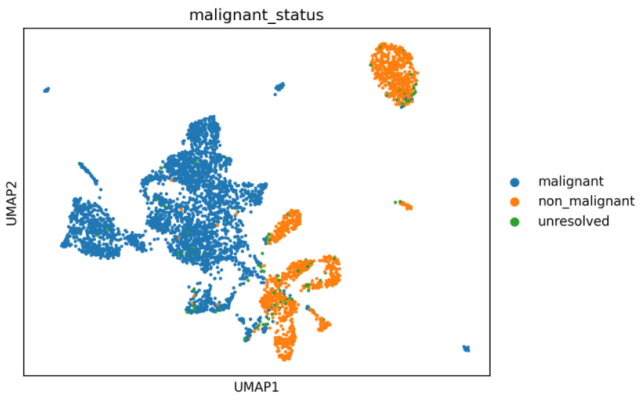


Figure 3. T-cell Exhaustion Landscape

Within the T-cell compartment, exhaustion scores varied substantially across the embedding, indicating that dysfunctional T-cell programs are spatially and transcriptionally heterogeneous rather than uniformly distributed.

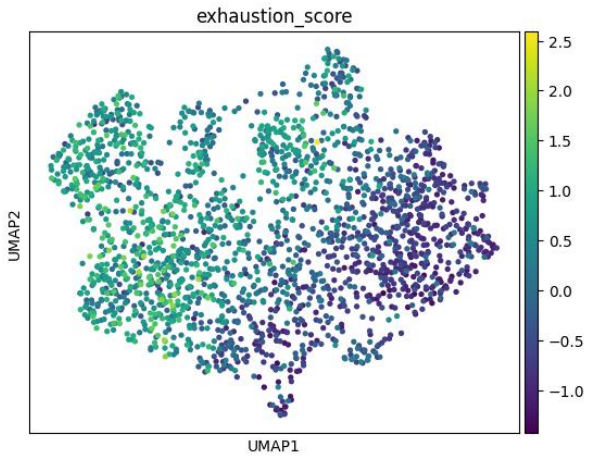
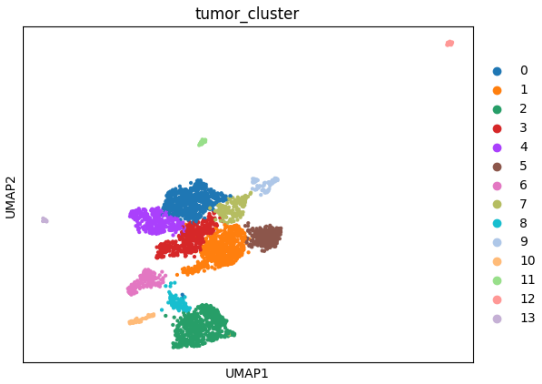


Figure 4. Tumour Subclustering

Reclustering of malignant-labeled cells revealed multiple tumour-associated subclusters, consistent with intratumoral heterogeneity and state diversification.



Discussion

This analysis demonstrates that melanoma consists of diverse immune and tumour states that can be resolved using scRNA-seq. The presence of multiple T-cell exhaustion states may have implications for immunotherapy responsiveness, while tumour plasticity states may contribute to resistance and progression.

An important practical finding was that metadata labels alone were insufficient for tumour-cell identification. Combining annotation, gene-signature scoring, marker analysis, and biological reasoning was necessary to obtain a cleaner tumour subset. This mirrors real-world single-cell workflows where iterative curation is often essential.

Limitations

- Exploratory analysis of a processed public dataset
- Limited clinical metadata available
- No batch correction or trajectory inference applied
- Functional states inferred from transcriptional signatures only

Future Directions

- Pseudotime trajectory analysis
- Cell-cell communication modeling
- Differential analysis between patients
- Integration with spatial transcriptomics
- Validation with external melanoma cohorts

Biological Significance

The biological implications of these findings are notable. First, heterogeneous exhaustion programs within T cells may help explain why some melanoma patients respond incompletely to checkpoint blockade despite the presence of infiltrating lymphocytes. Second, the coexistence of differentiated and plastic tumour states suggests adaptive capacity within the tumour compartment. Such plasticity has been associated in previous studies with metastasis, therapy escape, and reversible phenotype switching. Third, the presence of proliferative subpopulations indicates that only a fraction of tumour cells may actively drive expansion at a given time.

From a computational perspective, this project demonstrates the importance of combining unsupervised methods with domain knowledge. Clustering alone can identify structure, but marker genes and biological reasoning are required for meaningful interpretation.

Reproducibility and Code Availability

All analyses were performed in Jupyter notebooks using open-source Python tools. The workflow includes data loading, preprocessing, clustering, visualization, signature scoring, and downstream interpretation. The repository contains notebooks, exported figures, and project documentation to facilitate reuse and extension.

Conclusion

Single-cell analysis of melanoma revealed heterogeneous immune and tumour ecosystems, including exhaustion-associated T-cell states and multiple tumour transcriptional programs. This project highlights the utility of reproducible computational workflows for studying cancer biology and demonstrates practical skills in modern single-cell bioinformatics.

Author Statement

Independent portfolio project developed to build research-ready skills in cancer genomics, computational biology, and single-cell transcriptomics.